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(54) **REPLACEABLE, GAMMA STERILIZABLE
CORIOLIS FLOW SENSORS**

(71) Applicant: **Malema Engineering Corporation,**
Boca Raton, FL (US)

(72) Inventors: **Deepak Bhagwan Malani,** Boca Raton,
FL (US); **Jayasekar Rajagopalan,**
Cupertino, CA (US); **Claus**
Waldersdorff Knudsen, Fremont, CA
(US); **Renato De Castro,** Cupertino,
CA (US)

(73) Assignee: **Malema Engineering Corporation,**
Boca Raton, FL (US)

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CPC **G01F 1/8477** (2013.01)

(58) **Field of Classification Search**

CPC G01F 1/8409; G01F 1/8413; G01F 1/8418
See application file for complete search history.

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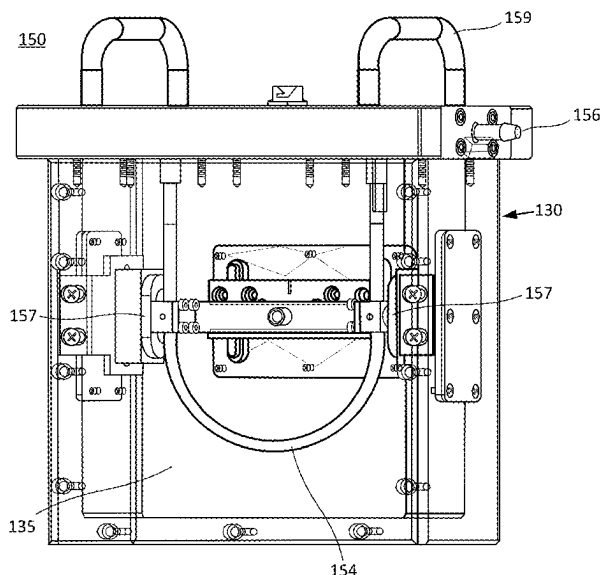
Primary Examiner — Erika J. Villaluna

(74) *Attorney, Agent, or Firm* — Fish & Richardson P.C.

(57) **ABSTRACT**

A Coriolis flow sensor includes a metal flow tube and an enclosure. The enclosure encloses the flow tube and is constructed at least partially from a gamma transparent material. The metal flow tube may be constructed from stainless steel. The gamma transparent material and the flow tube are thin enough to permit sterilization of an interior of the flow tube by gamma irradiation of the flow tube through the gamma transparent material. The enclosure is also shaped to facilitate locking and unlocking the Coriolis flow sensor in place on a mounting structure.

20 Claims, 11 Drawing Sheets



Related U.S. Application Data

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- (60) Provisional application No. 63/274,841, filed on Nov. 2, 2021.

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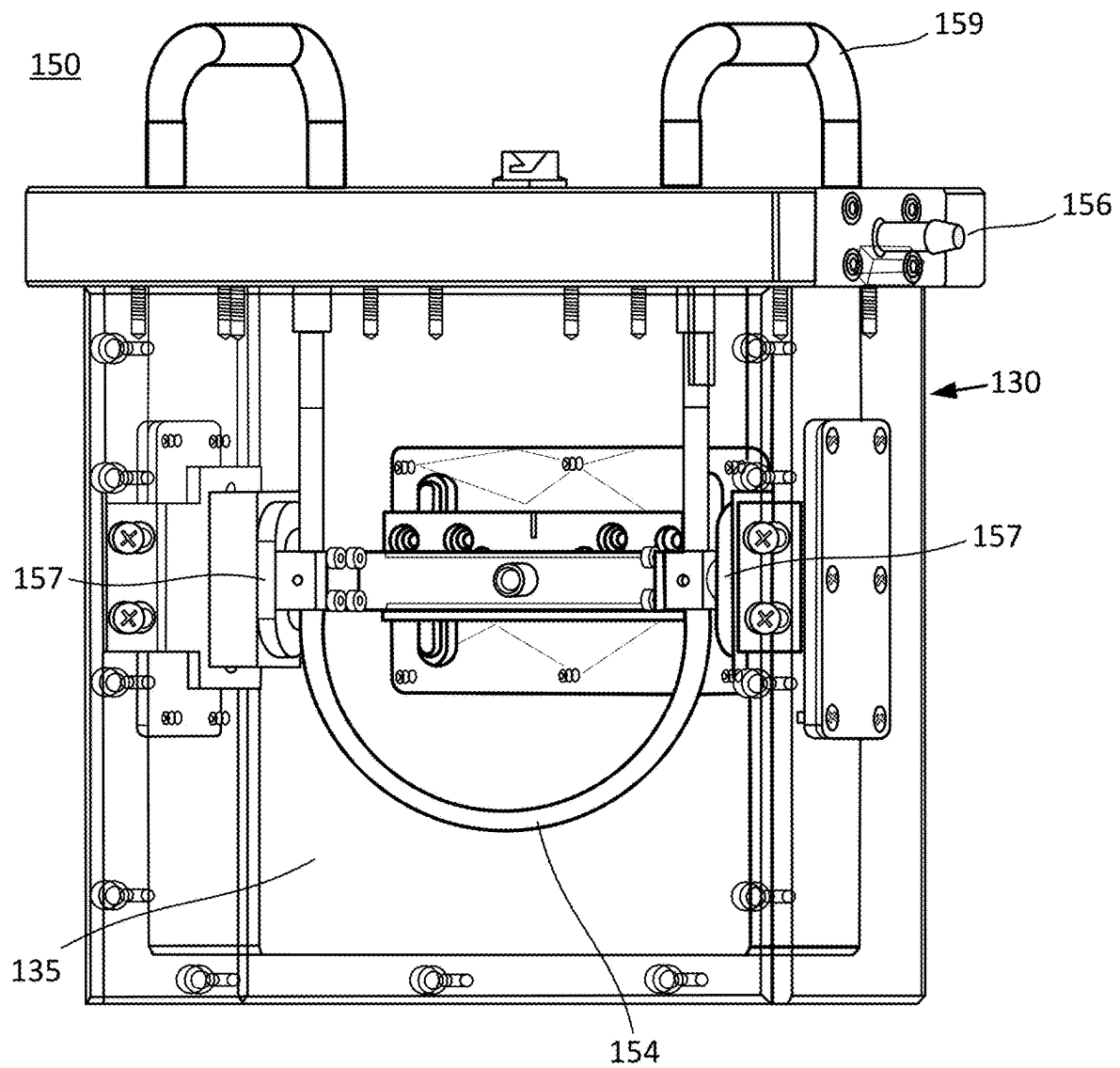


FIG. 1

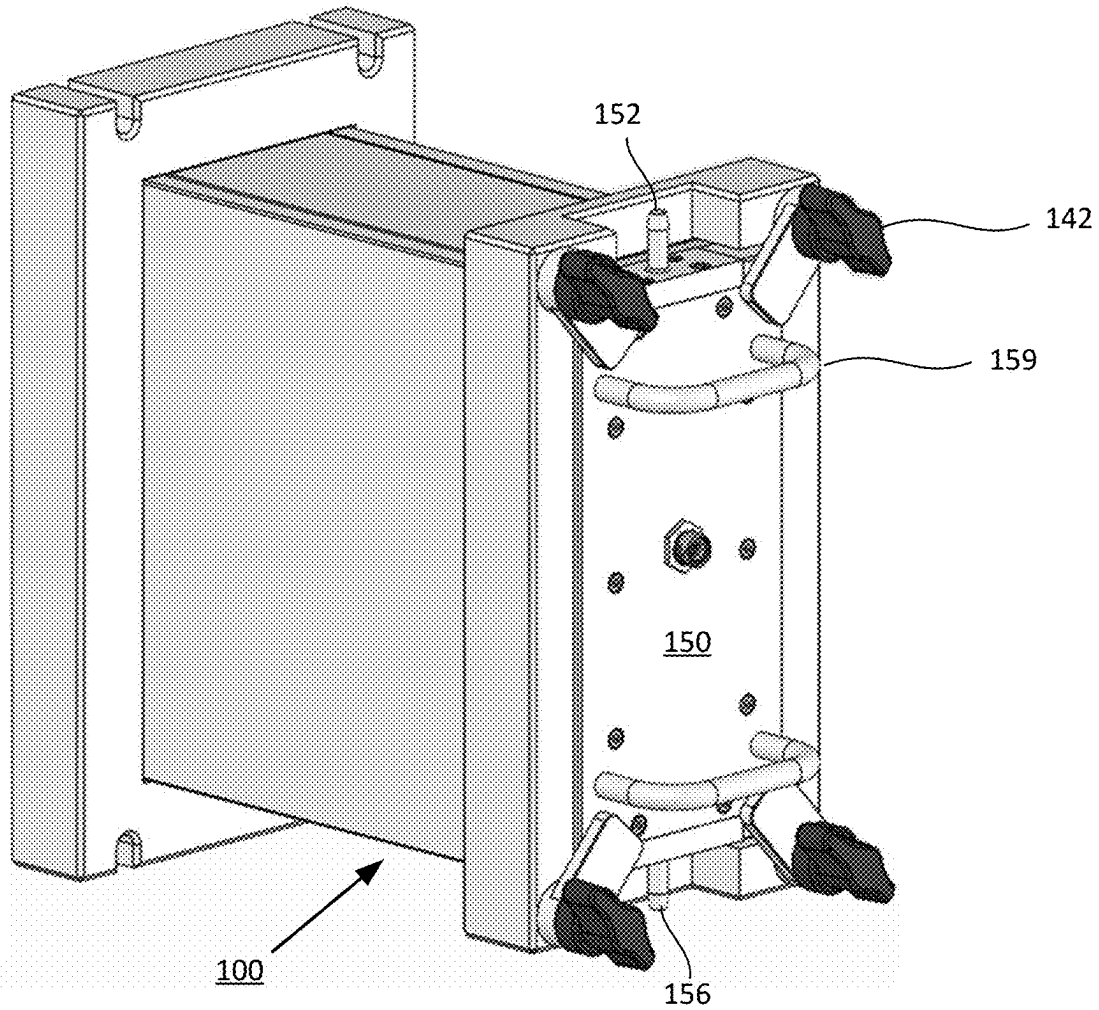


FIG. 2

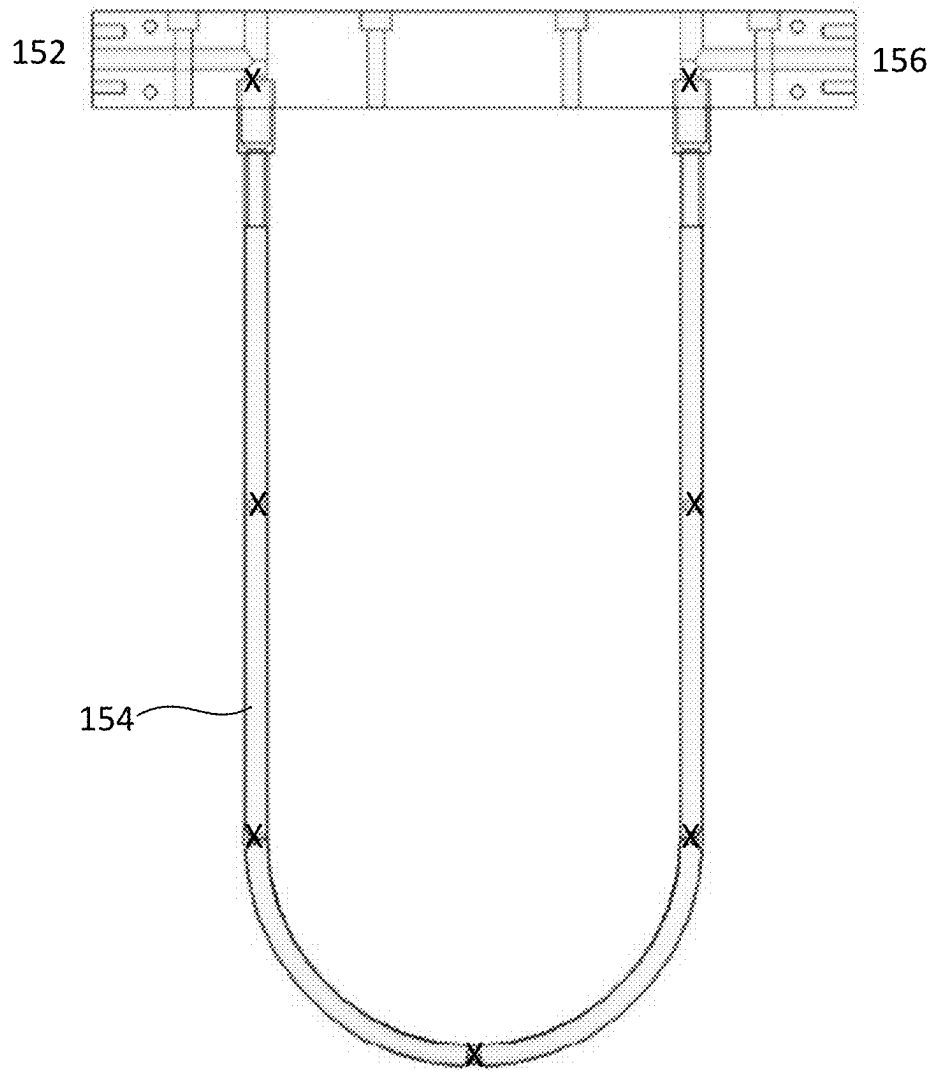
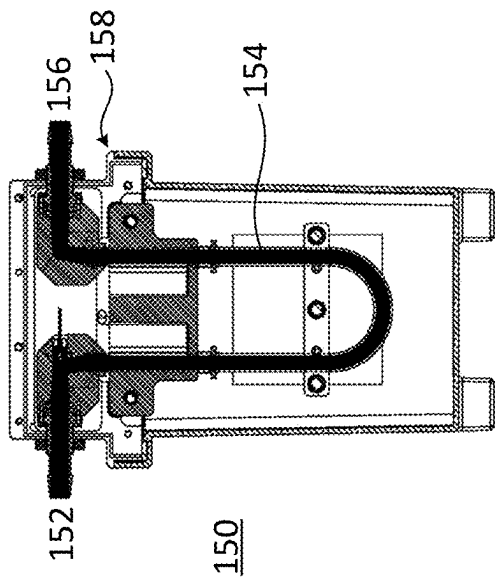
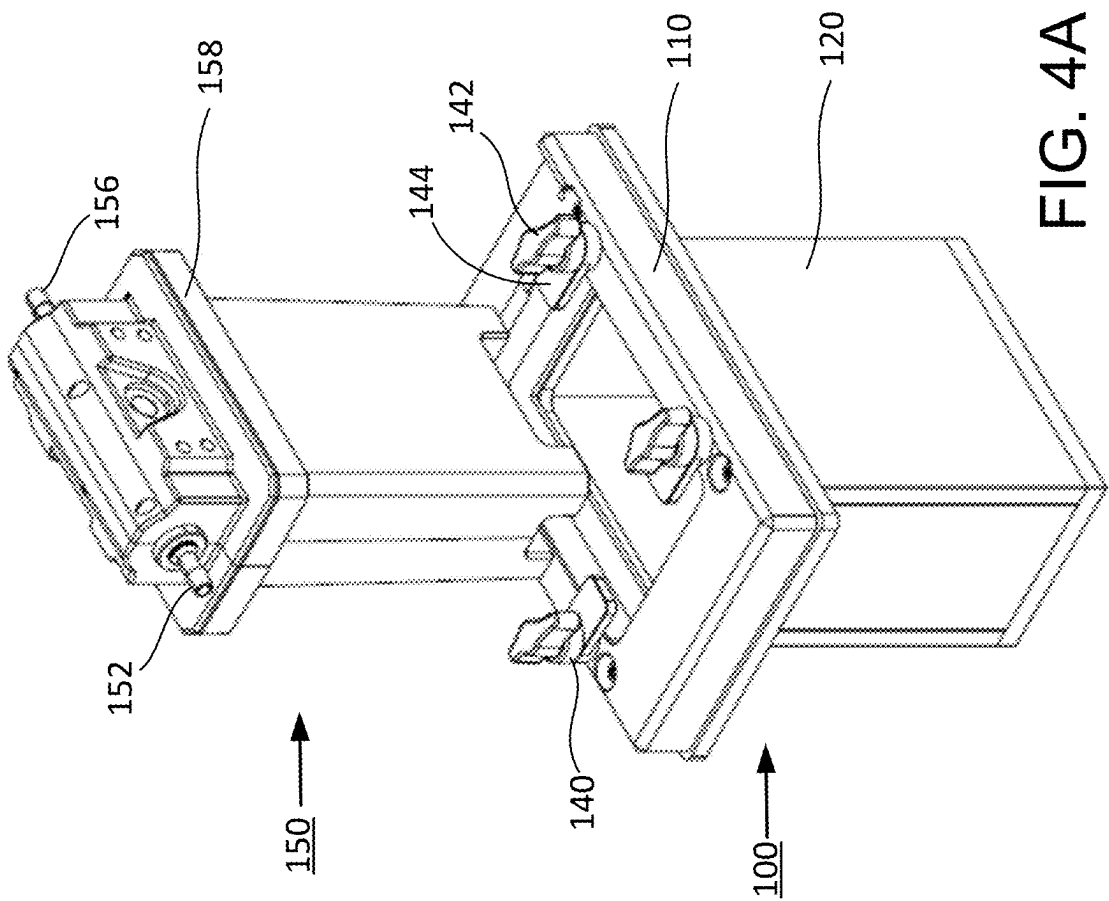
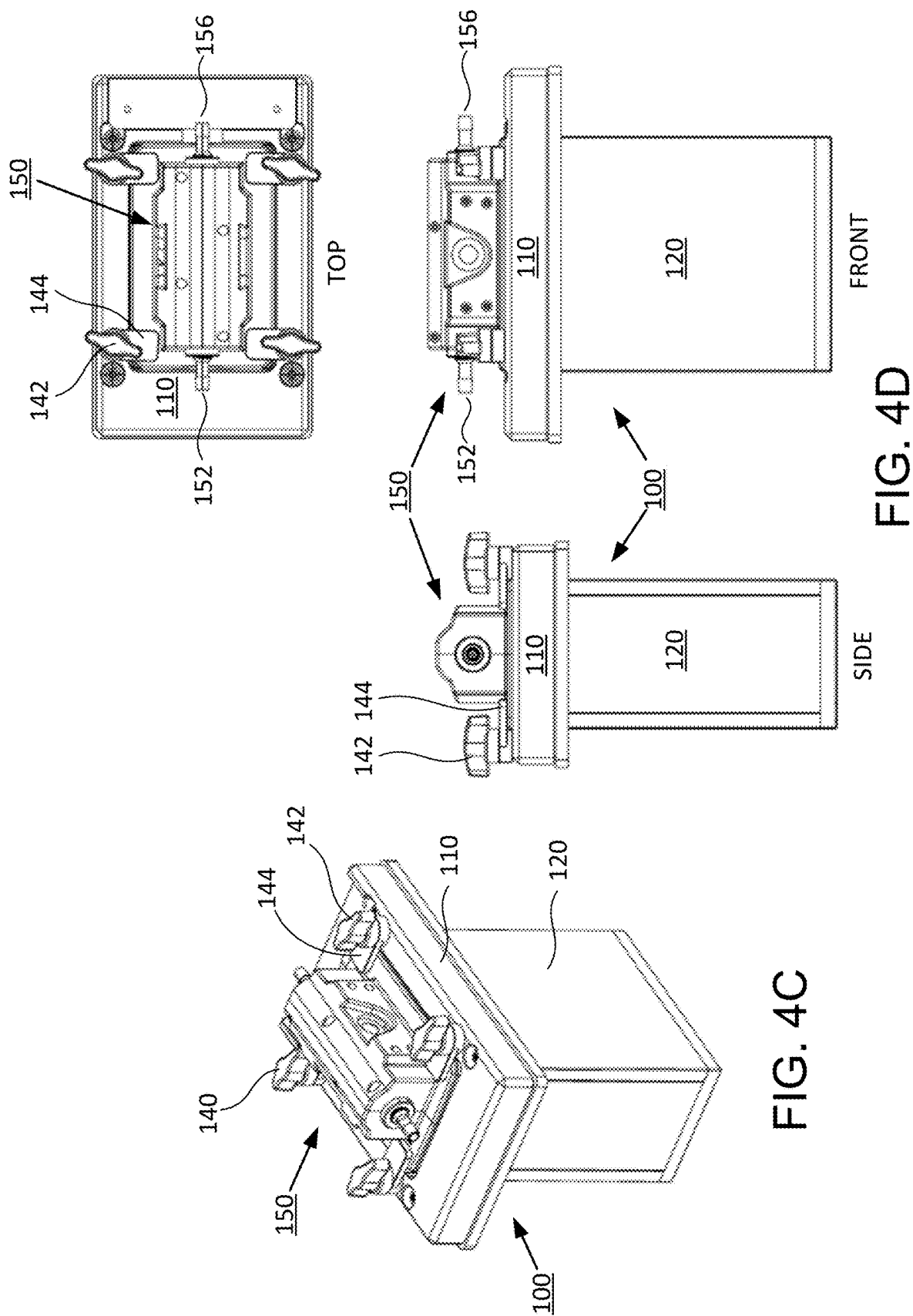


FIG. 3





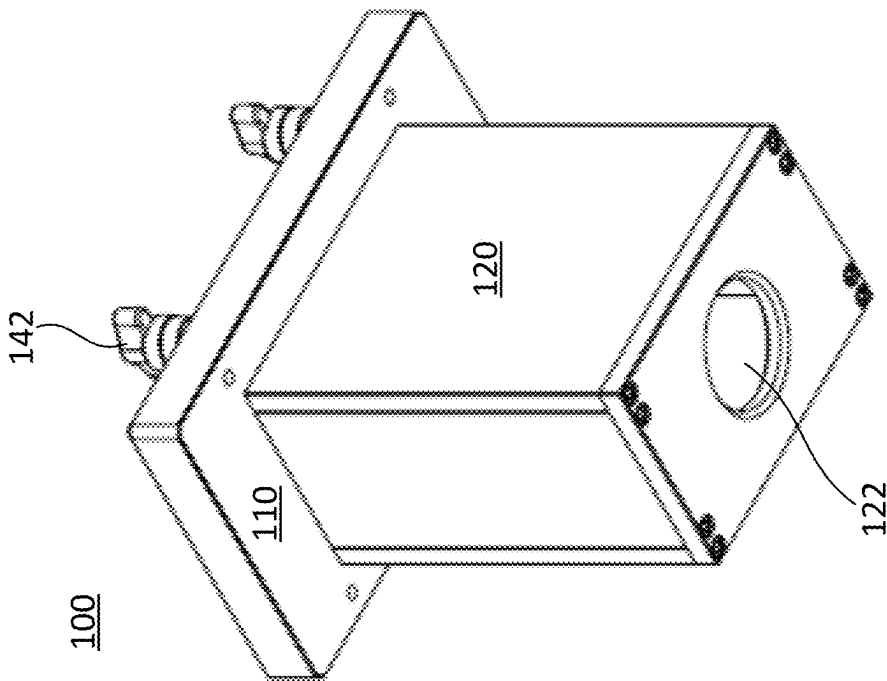


FIG. 5B

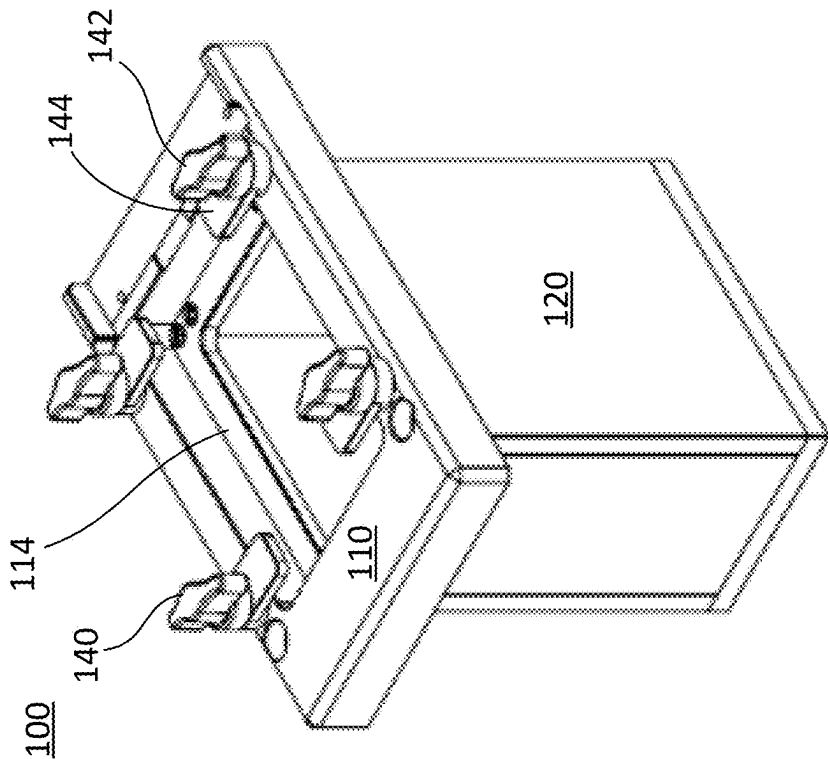


FIG. 5A

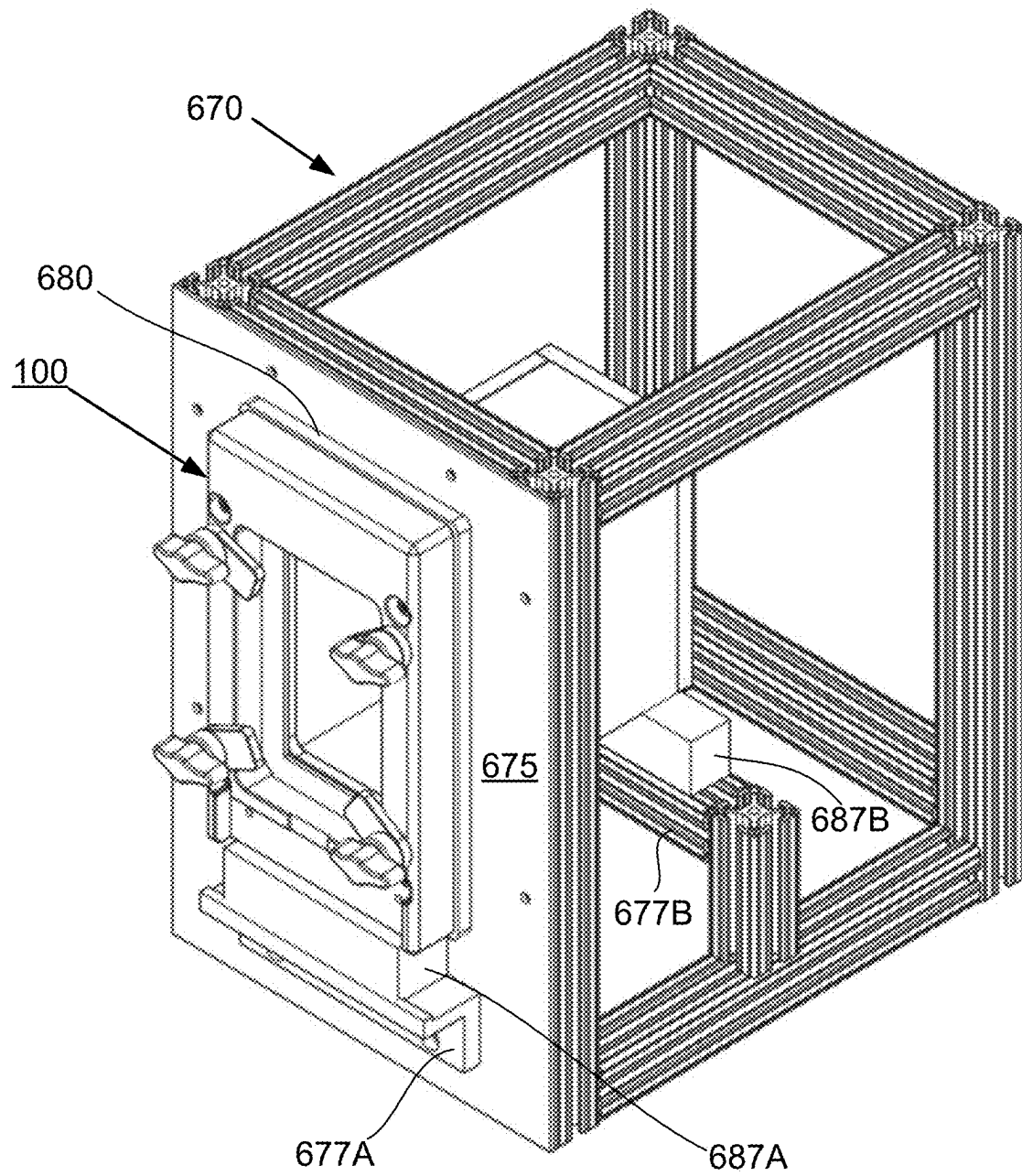
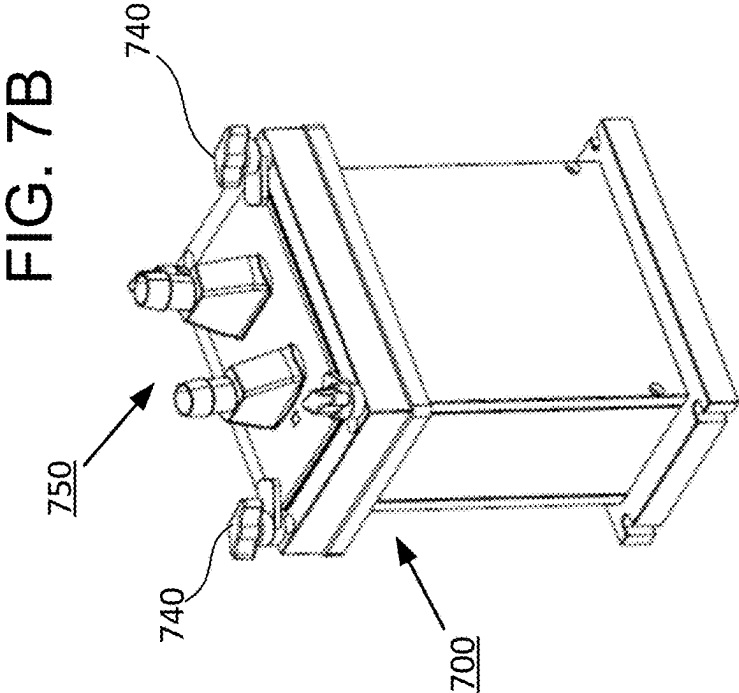
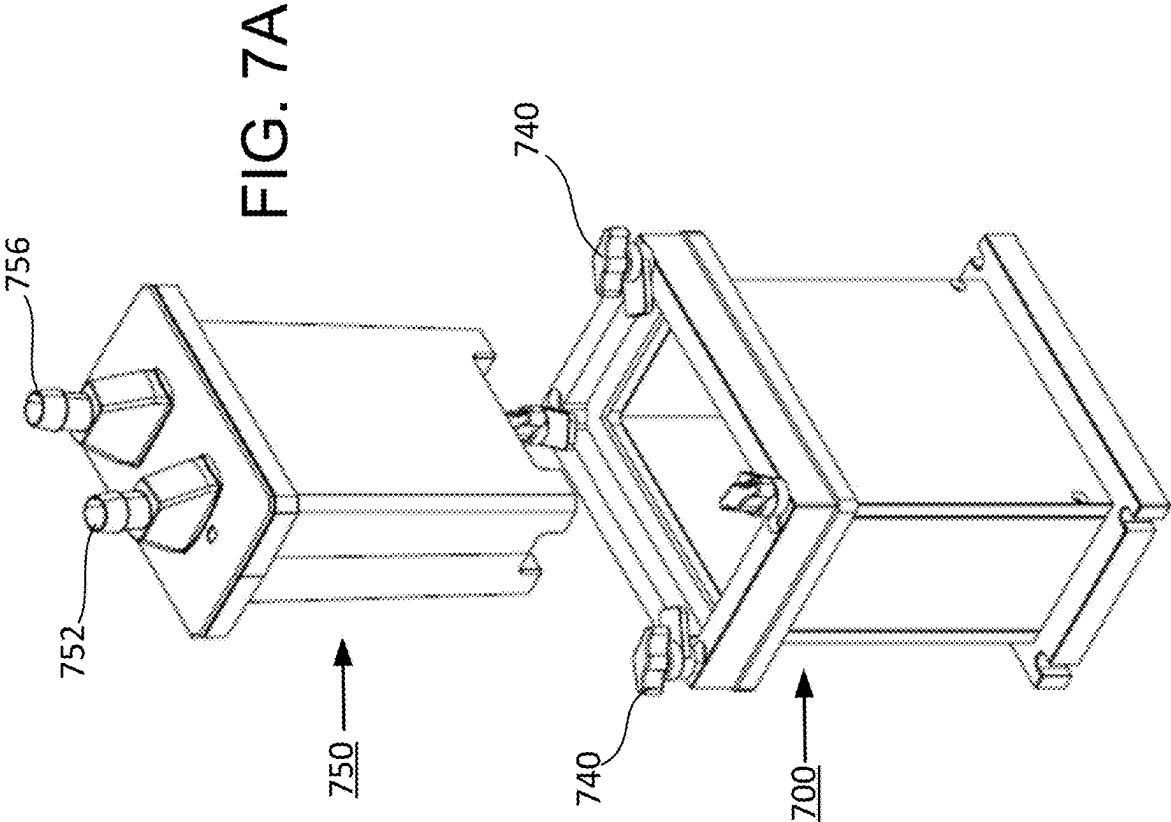


FIG. 6



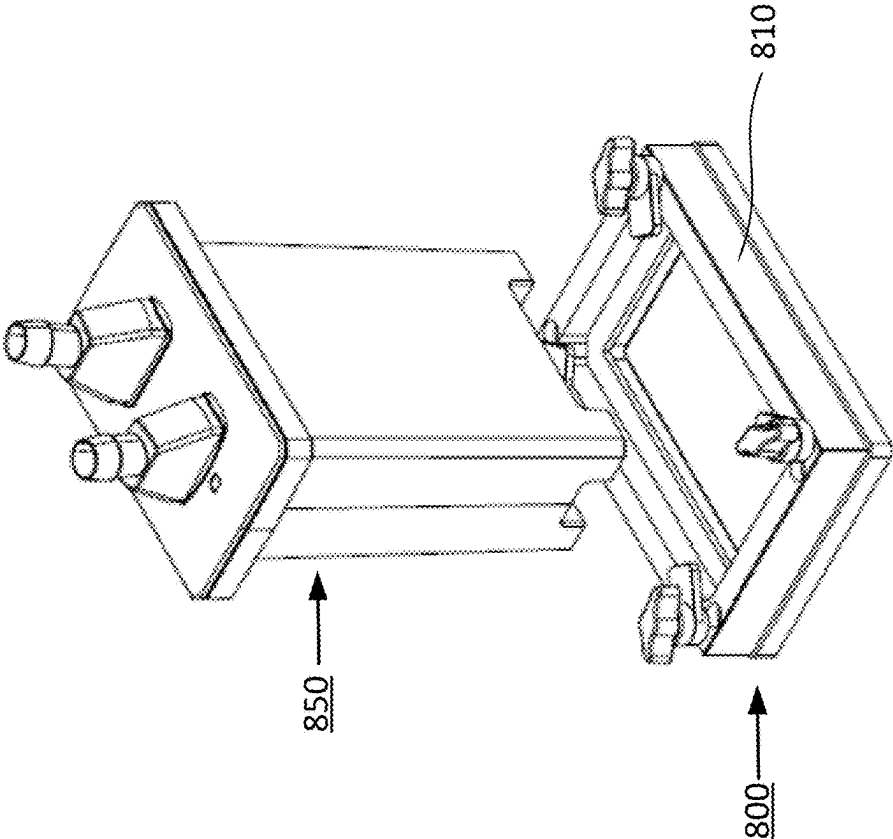


FIG. 8A

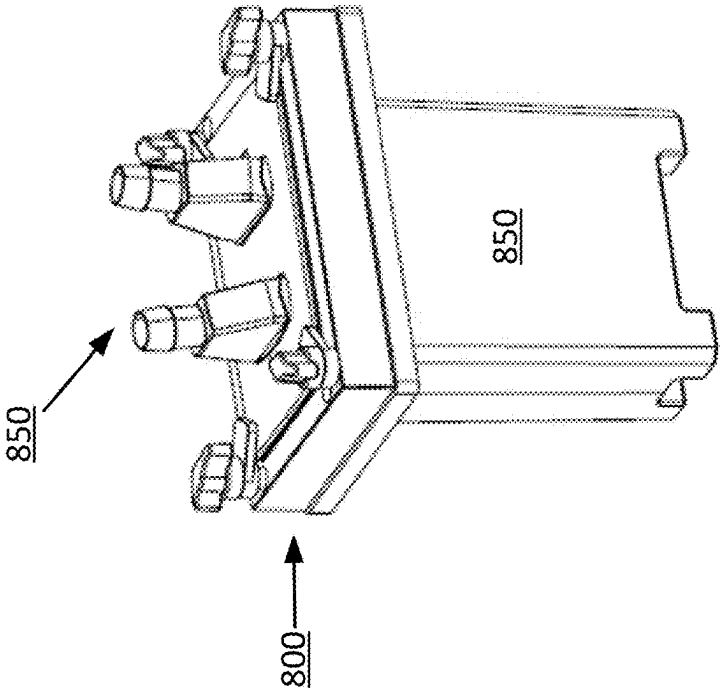


FIG. 8B

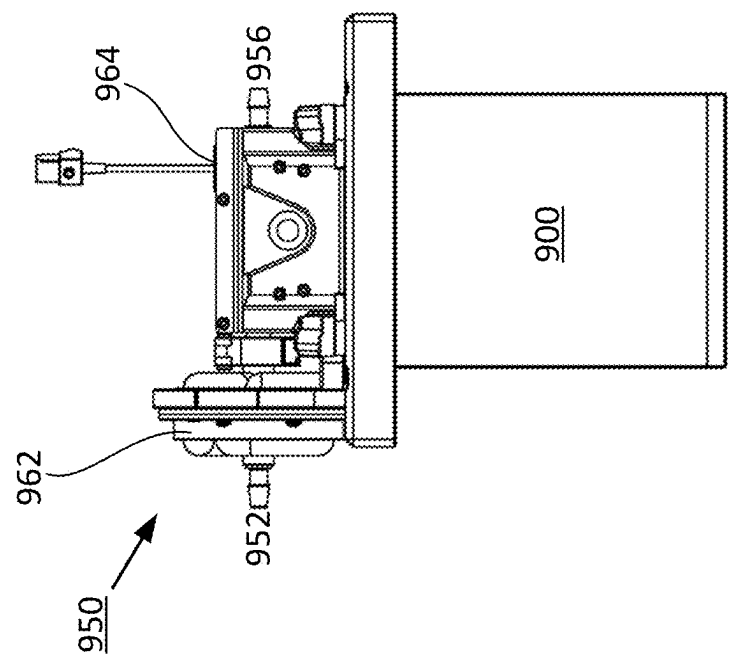


FIG. 9A

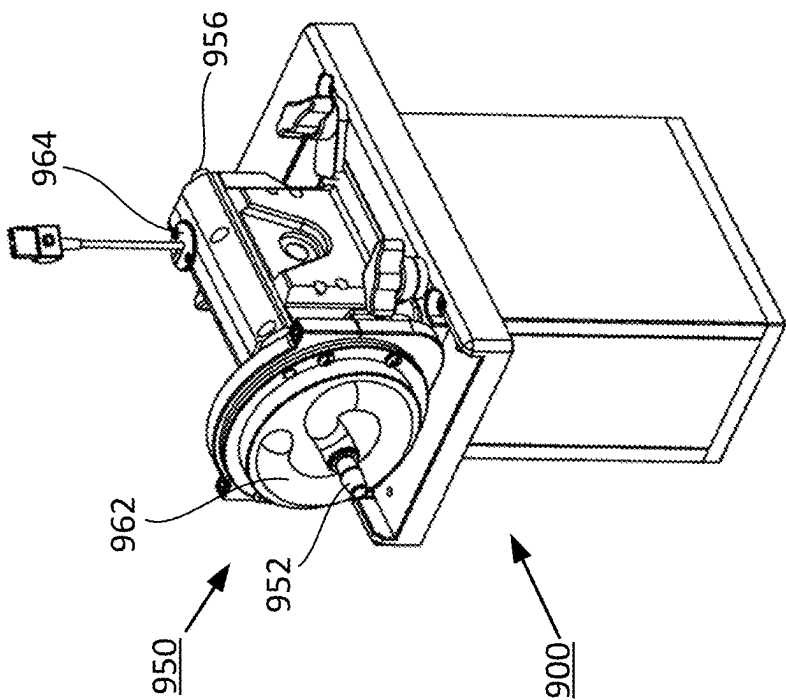


FIG. 9B

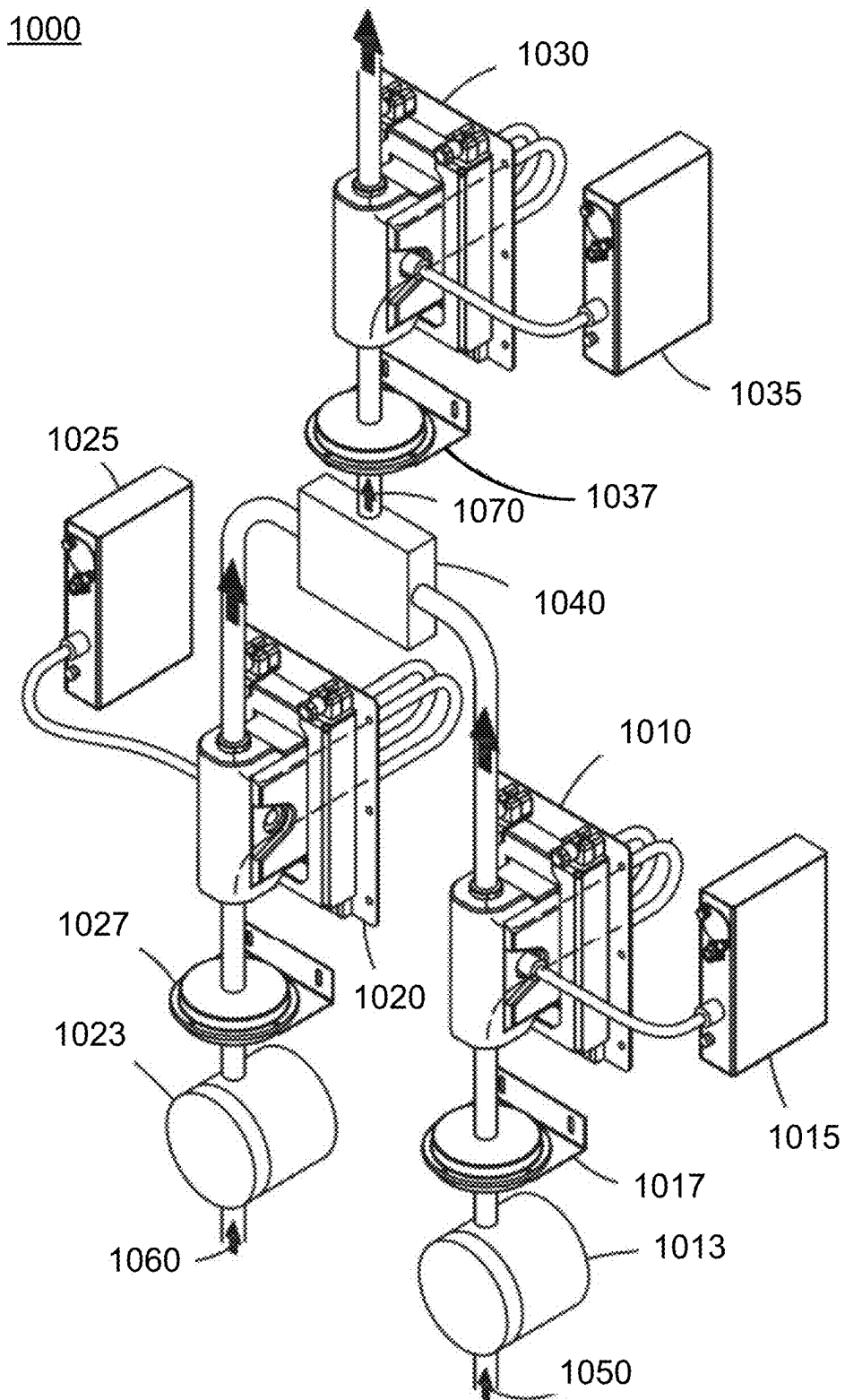


FIG. 10

1

REPLACEABLE, GAMMA STERILIZABLE CORIOLIS FLOW SENSORS

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a continuation of U.S. patent application Ser. No. 17/718,197, "Replaceable, Gamma Sterilizable Coriolis Flow Sensors," filed Apr. 11, 2022; which is a continuation-in-part of U.S. patent application Ser. No. 17/523,185, "Heavy Cradle For Replaceable Coriolis Flow Sensors," filed Nov. 10, 2021; which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application Ser. No. 63/274,841, filed Nov. 2, 2021. U.S. patent application Ser. No. 17/718,197 is also a continuation-in-part of U.S. patent application Ser. No. 17/702,554, "Coriolis Mass Flow Sensors Having Different Resonant Frequencies," filed Mar. 23, 2022; which is a continuation of U.S. patent application Ser. No. 16/846,061, "Coriolis Mass Flow Sensors Having Different Resonant Frequencies," filed Apr. 10, 2020, now U.S. Pat. No. 11,300,435. The subject matter of all of the foregoing is incorporated herein by reference in their entirety.

BACKGROUND

1. Technical Field

This disclosure relates generally to Coriolis flow sensors.

2. Description of Related Art

Many applications require the controlled flow of fluids. A flow process system usually includes a number of flow sensors to measure the flow rate of fluids. Coriolis flow sensors measure the flow rate of fluids based on vibrations caused by the Coriolis effect of fluid flowing through the sensor. However, in order to reduce cross-talk or destructive interference effects, conventional flow sensors may be attached to a large mass and the flow sensor itself may be constructed from heavy or thick materials.

However, these conventional designs can be expensive and are not suitable for single use or disposable applications. Also, sterilization of flow sensors having metal enclosures or metal components is typically implemented by using chemicals, which is not as effective and can cause cross-contamination of the flow sensors.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the disclosure have other advantages and features which will be more readily apparent from the following detailed description and the appended claims, when taken in conjunction with the examples in the accompanying drawings, in which:

FIG. 1 shows a Coriolis flow sensor.

FIG. 2 shows a Coriolis flow sensor inserted into a corresponding cradle.

FIG. 3 shows a test setup to verify sterilization of a Coriolis flow sensor using gamma irradiation.

FIG. 4A shows a perspective view of a Coriolis flow sensor and a corresponding cradle.

FIG. 4B shows a cross section view of the Coriolis flow sensor.

FIG. 4C shows a perspective view of the Coriolis flow sensor locked into the cradle.

2

FIG. 4D shows top, front and side views of the Coriolis flow sensor locked into the cradle.

FIGS. 5A and 5B show top perspective and bottom perspective views of the cradle.

FIG. 6 shows the cradle attached to a skid.

FIGS. 7A and 7B show perspective views of another embodiment of a Coriolis flow sensor and corresponding cradle.

FIGS. 8A and 8B show perspective views of yet another embodiment of a Coriolis flow sensor and corresponding cradle.

FIGS. 9A and 9B show perspective views of yet another embodiment of a Coriolis flow sensor and corresponding cradle.

FIG. 10 is a diagram of a flow measurement system containing Coriolis flow sensors.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

The figures and the following description relate to preferred embodiments by way of illustration only. It should be noted that from the following discussion, alternative embodiments of the structures and methods disclosed herein will be readily recognized as viable alternatives that may be employed without departing from the principles of what is claimed.

FIG. 1 shows a Coriolis flow sensor **150**. The Coriolis flow sensor **150** is a device that measures the flow rate of a fluid based on vibrations caused by the Coriolis effect of the fluid flowing through the sensor. The flow sensor **150** includes an inlet (hidden in FIG. 1), a flow tube **154** (or two or more flow tubes in some designs) and an outlet **156**. This provides a flow path for a fluid through the flow sensor **150**. The flow tubes **154** can vibrate, for example as driven by magnets and coils **157**. As the fluid flows through the flow tubes **154**, Coriolis forces produce a twisting vibration of the flow tubes, resulting in a phase shift in the vibration of the flow tubes. The fluid flow also changes the resonant frequency of the flow tubes. The flow sensor **150** includes transducers that generate electrical signals that are sensitive to the phase shift and/or change in resonant frequency. These signals may be processed to determine the mass fluid flow rate and/or density of the fluid.

It is desirable for the Coriolis flow sensor to have good accuracy over a broad operating range. As a result, the flow tubes may be constructed from materials with a high Young's modulus (tensile modulus), for example a Young's modulus of at least 50 GPa. The flow tubes may be metal flow tubes. They may be constructed from stainless steel, such as 316, 316L, 304 or 304L stainless steel. Other possible metals include Hastelloy, monel, nickel, titanium, and tantalum. Non-metals may also be used, such as zirconium. Table 1 tabulates Young's modulus for a number of different materials.

TABLE 1

Young's modulus for different materials	
Material	Young's modulus (GPa)
Steel, stainless 304, 304L, 316, or 316L	193
Tantalum	186
Titanium	116
Titanium alloy	110
Hastelloy	205

TABLE 1-continued

Young's modulus for different materials	
Material	Young's modulus (GPa)
Monel metal	179
Zirconium	94.5

The Coriolis flow sensor may have an operating range that ranges from 25 ml/min, or 20 ml/min or 15 ml/min or less on the low side, up to 4000 ml/min or 5000 ml/min or more on the high side, for 1/4" diameter flow tubes. Operating ranges will be different for different size flow tubes. 2" diameter flow tubes may have operating ranges up to 350 l/min or more. The operating range may be characterized by a turndown ratio, which is the ratio of the maximum flow rate divided by the minimum flow rate. The flow sensor may have an accuracy of $\pm 1\%$ of the actual flow rate over the operating range.

The Coriolis flow sensor is designed to be disposable or single use. Accordingly, the tubes 154 are enclosed in an enclosure 130. The enclosure 130 may be locked and unlocked in place on a mounting structure, as shown in FIG. 2. FIG. 2 shows a Coriolis flow sensor 150 locked into place in a cradle 100. Handles 159 may be used to insert the flow sensor 150 into the cradle 100, and to remove the flow sensor from the cradle. The inlet 152 and outlet 156 are visible in FIG. 2, but the flow tubes are not visible since they are within the cradle 100. In this example, the locking mechanism is four thumb screws 142. The enclosure 130 is shaped to slide into and out of the cradle 100.

The Coriolis flow sensor 150 is also designed to permit sterilization by gamma irradiation. Parts of the enclosure 130 are constructed from gamma transparent material, which are materials that have low attenuation of gamma irradiation. In FIG. 1, the side panels 135 are polycarbonate or other types of plastic. As a result, the interior of the flow tube 154 may be sterilized by gamma irradiation. The side panel 135 is thin enough and the walls of the flow tube 154 are thin enough to allow sufficient gamma irradiation to reach the inside of the flow tube. For example, the side panel 135 may be polycarbonate which readily passes gamma irradiation and the flow tubes may have a wall thickness not greater than 0.020". The wall thickness will vary depending on the size of the flow tubes. Table 2 below shows wall thicknesses for different size seamless stainless flow tubes. Maximum wall thicknesses may be approximately two times what is shown in Table 2.

TABLE 2

Wall thickness for different size flow tubes	
Outer diameter of flow tube (inches)	Wall thickness (inches)
1/4"	0.020
3/8"	0.028
1/2"	0.035
3/4"	0.035
1"	0.049
2"	0.065

The Coriolis flow sensor 150 also includes memory used to store calibration data or other types of data for the Coriolis flow sensor. If gamma sterilization is used, this memory may be gamma stable memory.

FIG. 3 shows a test setup to verify sterilization of a Coriolis flow sensor using gamma irradiation. For clarity, FIG. 3 shows only the flow tube 154 and not the enclosure 130. Gamma dosimeters are placed inside the flow tube 154 at the locations shown. The Coriolis flow sensor, including the plastic enclosure 130, is irradiated by gamma irradiation. The dosimeters indicate that sufficient radiation reaches the interior of the flow tube 154 to sterilize the flow sensor. The plastic walls of the enclosure 130 and walls of the flow tube 154 are thin enough to pass sufficient gamma irradiation.

In some designs, the Coriolis flow sensor may also be sterilized by other methods: ethylene oxide cleaning, sodium hydroxide cleaning or x-ray sterilization for example. If the Coriolis flow sensor is used multiple times (e.g., multi-use applications or continuous use applications), it may be sterilized between uses. To avoid having to remove the Coriolis flow sensor, it may be designed for clean-in-place processes or steam-in-place processes. Thus, the same flow sensor may be suitable for use in single-use applications, in multi-use applications and in continuous use applications.

The figures show examples of Coriolis flow sensors, but it should be understood that other types of Coriolis flow sensors may also be used. The number and shapes of tubes, the material and construction of the tubes and flow sensor, and the arrangement of the inlet and outlet may all be changed depending on the specific design of the Coriolis flow sensor. The Coriolis flow sensor may include one flow tube, or two or more flow tubes. The flow tubes may have different shapes in different flow sensors: a U-tube, a V-tube, an omega tube, or a straight tube, for example. Typically, Coriolis flow sensors are sized with connections from 1/16" to 2" hose barbs or tri-clamp fittings. Other types of fittings may also be used on Coriolis flow sensors. Typical flow ranges of these flow sensors range from 0.05 gm/min to 0.5 gm/min for the smallest (1/16" hose barb connections) size to 10 kg/min to 350 kg/min for the largest (2") size. Typical accuracies range from 0.1% to 1.0% of actual reading.

FIGS. 4-5 show different views of example embodiments of a Coriolis flow sensor 150 and corresponding cradle 100. FIG. 4 shows both the Coriolis flow sensor 150 and the cradle 100, where FIG. 4A is an exploded view, FIG. 4B shows just the flow sensor, FIG. 4C shows the assembled system, and FIG. 4D shows top, front and side views of the assembled system. FIGS. 5A and 5B are perspective views that show just the cradle 100 and locking mechanism 140.

The flow sensor 150 can be seen in cross section in FIG. 4B. The flow sensor 150 includes an inlet 152, a flow tube 154 (or two flow tubes in some designs) and an outlet 156. Because Coriolis flow sensors operate based on changes in the vibration of the flow tubes, vibration effects that are caused by sources other than the fluid flow may introduce inaccuracies. For example, if the flow sensor and other devices are mounted on a common support structure, then vibrations from pumps and other devices may mechanically couple to the flow sensor through the supporting structure. The vibration of the flow tubes may also be distorted or otherwise changed through resonant coupling to the surrounding support structure.

Zero drift is one such effect. Coriolis flow sensors are electrically powered on, even when they are not measuring flow. So when there is no flow being pumped or flowing through the Coriolis Flow tubes, the tubes continue to vibrate. Sometimes these tubes are empty and sometimes there is liquid in these tubes. Zero drift is a phenomenon which shows some minimal flow rate occurring when there is no real actual flow. One instance of zero drift is when there is dormant fluid left in the Coriolis flow tubes and a certain

amount of sloshing occurs. This minimal flow rate is very small and is usually a very small percentage of the minimum flow rate of each Coriolis flow sensor. In addition, vibrations from external mechanical devices such as pumps and valves also cause zero drift by interfering with the analog or digital output signal from a Coriolis flow sensor by contributing to it.

One way to reduce zero drift is to increase the mass of the flow sensor. More mass dampens out external mechanical vibrating interferences and also the sloshing of dormant liquid will be subdued due to heavier mass.

However, in some applications, the Coriolis flow sensors are not permanent. They are intended to be replaced fairly regularly. They may even be single use or considered to be disposable. Single use or disposable Coriolis flow sensors are used in the bio-pharmaceutical and pharmaceutical industries to manufacture vaccines including vaccines for Covid-19, active pharmaceutical ingredients for cell and gene therapy and nuclear medicine manufacturing.

In these cases, it is desirable to make the Coriolis flow sensor as lightweight and inexpensive as possible, so making a large and massive Coriolis flow sensor is not desirable. In addition, some applications may also require the sterilization of flow sensors. Gamma irradiation may be used to sterilize the flow sensor, in which case the flow sensor is constructed from materials that are gamma irradiatable, for example to a minimum of 25 kGy or 50 kGy or 65 kGy which may be the irradiation levels used for sterilization in certain bio-pharma applications.

In the examples shown herein, the effective mass of the Coriolis flow sensor **150** is increased by locking it to a heavy cradle **100** when it is in use. The cradle **100** has a mass that preferably is at least 10 to 30 times the mass of the Coriolis flow sensor. For example, typical Coriolis flow sensors may have masses in the range of 0.2 kg–3 kg and typical mass for the heavy cradle may then be 5 kg–80 kg.

The cradle **100** has a mounting structure **114** (see FIG. 5A) for the Coriolis flow sensor **100**, and a locking mechanism **140** is used to lock and unlock the Coriolis flow sensor in place on the mounting structure. The locking mechanism produces sufficient locking force when locked that the Coriolis flow sensor **150** and cradle **100** (as shown in FIG. 4A) vibrate together as a unitary body.

In the example of FIGS. 4-5, the cradle **100** includes a rectangular metal collar **110** which accounts for a significant amount of the mass of the cradle. The collar **110** has a rectangular aperture with an interior lip **114**, which is most visible in FIG. 5. The lip is also rectangular and annular in shape. The flow sensor **150** includes a plastic housing with a ridge **158**. The ridge **158** fits into the aperture of the metal collar **110** and presses against the lip **114** shown in FIG. 5A. The locking mechanism **140** applies force to the ridge **158** to hold the ridge rigidly against the lip **114**. The flow tubes **154** protrude through the annular opening in the lip **114**.

In this example, the locking mechanism **140** uses thumb screws **142** to create the force. When tightened, the thumb screws **142** apply pressure to tongues **144**, which in turn press the ridge **158** against the interior lip **114** of the metal collar **110**. The thumb screws are designed to apply a specific amount of force. In the example shown, the force is applied at four locking points arranged in a rectangular shape, although other arrangements are also possible. The applied force should be large enough to adequately reduce vibration of the flow sensor **150** relative to the collar **110**. As a result, the flow sensor **150** and cradle **100** will vibrate as a unitary body and the cradle **100** will effectively increase the mass of the flow sensor **150**, rather than the two vibrating

relative to each other. For example, each of the thumb screws **142** may apply 3 Newton-meters (Nm) of force or more, to hold the flow sensor **150** and cradle **100** rigidly relative to each other. This is an aggregate force of 12 Nm or more for all of the thumb screws. In other designs, lower locking forces may be acceptable, for example 10 Nm or more, or 5 Nm or more.

Applying uniform force is also important. Applying the same force at the four locking points allows for the pressure applied to be balanced. If the forces at the different locking points were not the same, the sensor would be imbalanced and the zero drift and resulting inaccuracy would be higher. In FIGS. 4-5, the same amount of force should be applied to each locking point. For example, the force applied at each of the locking points may be within 15% of each other, or more preferably within 10%, within 5% or even within 1% of each other.

One advantage of using thumb screws **142** is that the locking mechanism may be operated manually. The thumb screws **142** may be loosened, the tongues **144** rotated or swiveled away to release the flow sensor **150**, and the flow sensor removed and replaced with another flow sensor. This facilitates the replacement of flow sensors, including disposable and single use flow sensors. In some single use or disposable applications, the flow sensors may be removed and replaced in one minute or less.

The cradle **100** also includes enclosure **120**, which encloses the rest of the Coriolis flow sensor. The enclosure also adds mass. The enclosure shown in FIGS. 4-5 includes a cable hole **122** (see FIG. 5B) to allow power and data connections to the flow sensor for sensors with rear connecting cables. FIG. 1 shows a top mounting cable connection.

FIG. 6 shows the cradle **100** attached to a skid **670**. A skid is a mechanical framework on which equipment may be mounted. In this example, the cradle **100** is attached to a metal plate or panel **675**, which is attached to the skid **670**. A vibration dampening gasket **680** is positioned between the cradle **100** and the plate **675**. In the vertical direction, the cradle **100** is supported by cross members **677A** (an L bracket) and **677B** (a cross beam of the skid). Vibration dampening gaskets **687A** and **687B** are positioned between the cradle **100** and the cross members **677A** and **677B**.

Note that the heavy cradle **100** does not make direct contact with any part of the skid **670**. It is always separated by vibration gaskets **680**, **687**. The gaskets **680**, **687** provide vibration isolation between the cradle **100** and the skid **670** (and other components mounted on the skid). For example, the vibration gaskets may significantly dampen low frequency vibrations.

The heavy cradle **100** adds mass to the Coriolis flow sensor **150**, and the vibration gaskets **680**, **687** isolate the cradle and flow sensor from the rest of the flow process system. As a result, zero drift is reduced. For example, for smaller size sensors (e.g., tubing of 1/2 inch and less), zero drift was reduced from 100 g/min to 2.5 g/min. Typical minimum flow rate for these sensors is 500 g/min, so the zero drift is reduced to less than 1% of the minimum flow rate. For larger sensors (e.g., 3/4 and 1 inch tubing), zero drift was reduced from 200 g/min to 25 g/min. A typical minimum flow rate for these sensors is 6 kg/min, so the zero drift is reduced to less than 1% of the minimum flow rate.

FIGS. 7-9 show perspective views of additional embodiments of a Coriolis flow sensor **750**, **850**, **950** and corresponding cradle **700**, **800**, **900**. In FIG. 7, the flow sensor **750** has a vertical configuration, whereas the flow sensors in previous figures are in-line configurations. In an in-line

7

configuration, the inlet **152** and outlet **156** are in line with each other, but the flow typically is diverted in order to flow through the flow tubes. In a vertical configuration of FIG. 7, the inlet **752** and outlet **756** are not aligned with each other, but the flow is more in line with the flow tubes. The cradle and mounting structure may be designed to accommodate multiple different flow sensors, including both in-line Coriolis flow sensors and vertical Coriolis flow sensors. In addition, in FIG. 7, the locking points **740** are on the corners rather than along the sides.

In FIG. 8, the cradle **800** includes the collar **810** but does not have an enclosure. The flow sensor **850** protrudes through the collar **810** and is visible below the collar, as shown in FIG. 8B.

In FIG. 9, the Coriolis flow sensor has an in-line configuration with inlet **952** and outlet **956**. It also includes an integrated dampener **962** and integrated pressure sensor **964**. The dampener **962** is located on the inlet side of the flow sensor. The integrated dampener reduces vibrations in the fluid flow itself, for example as may be caused by a pulsating pump. Example dampeners are described in U.S. patent application Ser. No. 16/994,611 "Flow Dampener in Flow Measurement System," which is incorporated by reference in its entirety. Integrating the dampener and pressure sensor reduces the overall size and space requirement, compared to free-standing dampeners and pressure sensors that are connected to tubing on the inlet or outlet. It also reduces the amount of tubing required, which in turn reduces the amount of dead volume. Dead volume is the volume of fluid contained in tubing, sensors and other components, as this volume is lost and not converted to useable product when the system is flushed between batches. Reducing dead volume is important in pharmaceutical manufacturing, because dead volume is wasted product, which can be very valuable. The integrated pressure sensor can also produce more accurate pressure readings for calibrating the Coriolis flow sensor, since it is measuring pressure closer to the actual flow tubes.

FIG. 10 is a diagram illustrating a flow measurement system **1000** that uses single-use or disposable flow sensors **1010**, **1020**, and **1030**, as described above. The flow measurement system **1000** also includes two pumps **1013** and **1023**, three controllers **1015**, **1025**, and **1035**, three dampeners **1017**, **1027**, and **1037**, and a mixing manifold **1040**. The flow measurement system **1000** can be a part of a process skid, e.g., a biopharmaceutical or pharmaceutical skid. The pumps **1013**, **1023** preferably are located close to the flow sensors **1010**, **1020** and in-line with the flow sensors. The dampeners **1017**, **1027** are used to mitigate cross-talk from the pumps **1013**, **1023** to the flow sensors **1010**, **1020**, although they may not be required depending on the application.

Two fluids **1050** and **1060** enter the flow measurement system **1000**. One fluid **1050** enters the pump **1013**, which pumps the fluid through the dampener **1017** and flow sensor **1010** into the mixing manifold **1040**. The other fluid **1060** enters the pump **1023**, which pumps the fluid **1060** through the dampener **1027** and flow sensor **1020** into the mixing manifold **1040**. The flow sensors **1010**, **1020** measure flow characteristics (e.g., mass flow rate, volumetric flow rate, flow density, etc.) of the fluids **1050** and **1060**, respectively.

The two fluids **1050**, **1060** are combined at the mixing manifold **1040**, and the mixture **1070** of the two fluids flows out of the manifold. In some embodiments, the mixing manifold **1040** also receives other fluids or matter that can be mixed or reacted with fluids **1050** and **1060** to produce fluid **1070**. The mixing manifold **1040** may include a pump

8

that pumps the fluid **1070** through the dampener **1037** and flow sensor **1030**. The flow sensor **1030** measures flow characteristics of fluid **1070**.

The flow sensors **1010**, **1020**, **1030** can operate simultaneously. In some embodiments, at least two of the flow sensors **1010**, **1020**, and **1030** have similar operating ranges. The controllers **1015**, **1025**, **1035** receive signals from the flow sensors **1010**, **1020**, **1030** and conducts flow analysis based on the signals. The flow analysis includes, for example, determination of flow rate based on signals representing phase shift of the flow tubes, determination of flow density based on signals representing change in resonant frequency of the flow tubes, detection of bubbles in the fluids based on change in flow density, and determination of other flow characteristics of the fluids.

Although the detailed description contains many specifics, these should not be construed as limiting the scope of the invention but merely as illustrating different examples. It should be appreciated that the scope of the disclosure includes other embodiments not discussed in detail above. Various other modifications, changes and variations which will be apparent to those skilled in the art may be made in the arrangement, operation and details of the method and apparatus disclosed herein without departing from the spirit and scope as defined in the appended claims. Therefore, the scope of the invention should be determined by the appended claims and their legal equivalents.

What is claimed is:

1. A process comprising:

obtaining a Coriolis mass flow sensor comprising at least one metal flow tube enclosed within an enclosure that is shaped to facilitate locking and unlocking the Coriolis flow sensor in place on a mounting structure, the enclosure constructed at least partially from a gamma transparent material, wherein the gamma transparent material and the metal flow tube are thin enough to permit sterilization of an interior of the metal flow tube by gamma irradiation of the metal flow tube through the gamma transparent material;

sterilizing the Coriolis mass flow sensor by using Gamma irradiation; and

storing calibration data for the Coriolis flow sensor in a gamma stable memory.

2. The process of claim 1 further comprising locking, by a locking mechanism, the Coriolis mass flow sensor in place on the mounting structure of a cradle, the cradle having a mass of at least ten (10) times a mass of the Coriolis mass flow sensor, wherein the locking mechanism produces sufficient locking force that the Coriolis flow sensor and the cradle vibrate as a unitary body.

3. The process of claim 2 further comprising unlocking the locking mechanism to release the Coriolis mass flow sensor.

4. The process of claim 2, wherein the locking mechanism comprises thumb screws and tongues, the tongues arranged to overlap a ridge of the Coriolis mass flow sensor and, with the thumb screws tightened, press the ridge against an interior lip of a metal collar of the cradle.

5. The process of claim 2, wherein the Coriolis flow sensor has a zero drift of not more than 1% of a minimum flow rate measured by the Coriolis flow sensor.

6. The process of claim 3 further comprising:

removing the Coriolis mass flow sensor from the cradle; and

replacing the Coriolis mass flow sensor with a different Coriolis mass flow sensor.

7. The process of claim 1, wherein the at least one flow tube is constructed from stainless steel.

8. The process of claim 1, wherein the gamma transparent material is plastic or polycarbonate.

9. The process of claim 1, wherein the at least one flow tube is a U-tube, a V-tube, an Ω -shaped tube, or a straight tube.

10. A Coriolis flow sensor comprising:

at least one metal flow tube enclosed within an enclosure that is shaped to facilitate locking and unlocking the Coriolis flow sensor in place on a mounting structure, the enclosure constructed at least partially from a gamma transparent material, wherein the gamma transparent material and the metal flow tube are thin enough to permit sterilization of an interior of the metal flow tube by gamma irradiation of the metal flow tube through the gamma transparent material.

11. The Coriolis flow sensor of claim 10 further comprising a gamma stable memory that stores calibration data for the Coriolis flow sensor.

12. The Coriolis flow sensor of claim 10, wherein the Coriolis flow sensor has an accuracy of $\pm 1\%$ of a flow rate over an operating range that has a turndown ratio of maximum flow rate to minimum flow rate of at least 200.

13. The Coriolis flow sensor of claim 10, wherein the at least one flow tube is a metal flow tube.

14. The Coriolis flow sensor of claim 13, wherein the at least one metal flow tube is constructed from a stainless steel.

15. The Coriolis flow sensor of claim 10 further comprising:

a locking mechanism configured to lock the Coriolis flow sensor in place on the mounting structure of a cradle, the cradle having a mass of at least ten (10) times a mass of the Coriolis flow sensor, wherein the locking mechanism produces sufficient locking force that the Coriolis flow sensor and the cradle vibrate as a unitary body.

16. The Coriolis flow sensor of claim 15, wherein the locking mechanism comprises thumb screws and tongues, the tongues arranged to overlap a ridge of the Coriolis flow sensor and, with the thumb screws tightened, press the ridge against an interior lip of a metal collar of the cradle.

17. The Coriolis flow sensor of claim 10, wherein the at least one flow tube is a U-tube, a V-tube, an Ω -shaped tube, or a straight tube.

18. The Coriolis flow sensor of claim 10, wherein the Coriolis flow sensor is a disposable, single-use flow sensor.

19. A flow process system comprising:

a skid;

a cradle mounted on the skid, the cradle comprising a mounting structure;

a Coriolis flow sensor installed in the mounting structure, the Coriolis flow sensor comprising at least one metal flow tube enclosed within an enclosure that is shaped to facilitate locking and unlocking the Coriolis flow sensor in place on the mounting structure, the enclosure constructed at least partially from a gamma transparent material, wherein the gamma transparent material and the metal flow tube are thin enough to permit sterilization of an interior of the metal flow tube by gamma irradiation of the metal flow tube through the gamma transparent material; and

a locking mechanism that locks the Coriolis flow sensor in place on the mounting structure.

20. A flow process system comprising:

a skid;

a mixing manifold mounted on the skid;

a first flow path comprising a first pump and a first Coriolis flow sensor positioned along the first flow path and mounted on the skid, wherein the first pump and first Coriolis flow sensor are in fluid communication with a first inlet of the mixing manifold; and

a second flow path comprising a second pump and a second Coriolis flow sensor positioned along the second flow path and mounted on the skid, wherein the second pump and second Coriolis flow sensor are in fluid communication with a second inlet of the mixing manifold,

wherein each of the Coriolis flow sensors comprises at least one flow tube enclosed within an enclosure that is shaped to facilitate locking and unlocking the Coriolis flow sensor in place on a mounting structure, the enclosure constructed at least partially from a gamma transparent material, wherein the gamma transparent material and the flow tube are thin enough to permit sterilization of an interior of the flow tube by gamma irradiation of the flow tube through the gamma transparent material.

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